DEVASHISH SARVADE

Data Engineer

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PROFESSIONAL SUMMARY:

Results-driven Senior Data Engineer with 5+ years of experience architecting scalable, AI-integrated data platforms across AWS, Azure, and Databricks ecosystems. Proven expertise in building high-performance ETL/ELT pipelines using Apache Spark, Kafka, Airflow, Snowflake, Synapse, and Redshift for large-scale, real-time, and batch data processing. Skilled in operationalizing machine learning models for use cases such as fraud detection, anomaly detection, sentiment analysis, and forecasting using SageMaker, Azure ML, and Databricks MLflow. Adept at embedding AI workflows into production pipelines and visualizing insights through Tableau, Power BI, and SQL-based transformations. Strong background in data governance, quality (Great Expectations, dbt), CI/CD automation (Terraform, DevOps, CodePipeline), and infrastructure as code, with a focus on delivering reliable, compliant, and scalable data solutions in Agile, cross-functional environments.

EXPERIENCE:

United Airlines — ***Houston, TX***

Sr. Data Engineer — ***Jul '24 – Present***

- Designed and deployed robust, scalable ETL/ELT pipelines using AWS Glue, Amazon S3, and AWS Lambda, integrated with Databricks to process structured and semi-structured data from diverse sources at scale.
- Built and maintained PySpark-based data transformation workflows on Databricks, optimizing job execution using caching, broadcast joins, and partitioning to accelerate analytics workloads.
- Architected data lakehouse solutions by integrating Databricks Delta Lake with Snowflake, enabling unified data processing and analytics across real-time and historical datasets.
- Developed and managed real-time data ingestion pipelines using Apache Kafka, AWS Kinesis, and Databricks Structured Streaming for operational telemetry, user clickstream, and IoT data processing.
- Designed high-performance Snowflake data models using star and snowflake schemas, clustering keys, and materialized views to support fast querying for BI and ML use cases.
- Integrated AI/ML models into pipelines using SageMaker and Databricks MLflow, automating workflows for fraud detection, demand forecasting, and predictive maintenance across aviation operations.
- Orchestrated model training, validation, and inference pipelines within Databricks, using Feature Store for consistent feature reuse and automated deployment of models to production environments.
- Created natural language processing (NLP) pipelines for analyzing customer service interactions and feedback using sentiment analysis and topic modeling, with results stored in Snowflake and visualized via Tableau.
- Automated pipeline orchestration and monitoring using Apache Airflow and Databricks Workflows, ensuring job scheduling, retry logic, SLA tracking, and alerting through integrated Slack and CloudWatch.
- Built CI/CD pipelines with Terraform,AWS CodePipeline, and GitHub Actions to manage infrastructure as code and automate deployment of Databricks notebooks, jobs, and Snowflake scripts.
- Implemented data quality checks using Great Expectations within Databricks notebooks, validating schema compliance, null thresholds, and value ranges before persisting data to Snowflake.
- Developed AI-powered anomaly detection pipelines using unsupervised learning models in Databricks, flagging abnormal flight delays, sensor readings, and ticket transactions in near real-time.
- Conducted performance tuning of Snowflake queries, employing result caching, proper clustering, and query profiling to reduce cost and improve dashboard load times by 40%.
- Enforced enterprise-grade data governance using AWS IAM, Lake Formation, and Snowflake's RBAC policies to ensure data security, access control, and regulatory compliance.
- Partnered with Data Science and Analytics teams to operationalize AI models, embedding them within production pipelines and feeding results into business intelligence platforms like Tableau.
- Led an internal initiative to build a real-time AI insights dashboard, integrating multiple pipeline outputs to enable actionable decision-making by operations and customer service teams.
- Created detailed data monitoring dashboards using Tableau and Python-based visualization tools (Dash, Matplotlib), providing key performance insights into pipeline execution, error tracking, and data health metrics for business stakeholders and engineers.

Environment: AWS (S3, Glue, Lambda, Kinesis, IAM, SageMaker, CloudWatch), Databricks (PySpark, Delta Lake, MLflow, Feature Store), Snowflake, Kafka, Airflow, Terraform, Great Expectations, Python, SQL, Tableau, GitHub Actions, Agile.

CapitalOne — ***Plano,TX***

Data Engineer — ***Sep '23 – Jun '24***

- Developed and maintained scalable ETL/ELT pipelines on Azure Databricks using PySpark and Delta Lake, processing high-volume financial transaction data for analytical and machine learning purposes.
- Designed and implemented batch and streaming data pipelines using Azure Synapse Analytics, Azure Data Factory (ADF), and Azure Event Hubs, supporting near real-time fraud detection and credit risk assessment use cases.
- Built AI-integrated feature engineering pipelines using Databricks ML and PySpark to support fraud detection, customer segmentation, and personalized credit recommendations.
- Operationalized machine learning models trained in Azure Databricks by integrating them into production data flows with MLflow,

enabling version-controlled, reproducible AI workflows.

- Migrated legacy Hadoop workloads and on-prem SQL Server systems to Azure Synapse and Data Lake Gen2, reducing infrastructure cost and improving performance scalability.
- Created and optimized complex SQL queries and views (window functions, CTEs, partitioning strategies) in Azure Synapse, Snowflake, and Azure SQL Database to support analytics and dashboarding needs.
- Built real-time streaming pipelines using Event Hubs and Structured Streaming in Databricks, consuming millions of transactions and flagging anomalies using pre-trained ML models.
- Developed NLP pipelines for analyzing customer feedback and chatbot transcripts using Databricks, with output stored in Synapse for downstream reporting and operational insight.
- Automated workflow orchestration using Apache Airflow and Azure Data Factory pipelines, ensuring efficient scheduling, retries, and SLA enforcement for end-to-end data workflows.
- Implemented CI/CD for data and ML pipelines using Terraform, Azure DevOps, and Databricks Repos, ensuring consistent environment management and release governance.
- Enforced robust data security and governance practices using Azure Key Vault, RBAC, data masking, and sensitivity labels within Synapse and ADLS to meet enterprise compliance standards.
- Designed high-performance, scalable Synapse data models with partitioned fact tables, materialized views, and query optimization for real-time executive dashboards and reporting tools.
- Developed Power BI dashboards by connecting to Azure Synapse and Snowflake, integrating ML-driven metrics such as fraud scores, sentiment classification, and predicted credit limits.
- Integrated Great Expectations and dbt with Databricks workflows to perform rigorous data quality validations before loading into Synapse, improving trust in production data assets.
- Collaborated with data scientists and business analysts in Agile sprints to deploy new AI features, improve data accessibility, and support data-driven financial services initiatives.
- Designed scalable and high-performance data warehouse solutions using Star and Snowflake schemas in Snowflake, Synapse, and Big Query, supporting advanced analytics workloads.
- Integrated Power BI and Tableau dashboards with cloud data sources, enabling real-time business intelligence and data-driven decision-making.
- Worked in Agile development sprints, collaborating with cross-functional teams to optimize data pipeline performance and implement new features.

Environment: Azure (Databricks, SynapseAnalytics, Data Factory, Event Hubs, Data Lake Gen2, Key Vault,Azure DevOps), PySpark, MLflow, SQL (Azure SQL, Synapse, Snowflake, PostgreSQL), Apache Airflow, dbt, Terraform, Python, NLP, Power BI, Agile.

Tata Consultancy Services — ***Hyderabad, India***
Data Engineer — ***Aug '19 - Apr '23***

- Engineered robust big data pipelines using Apache Spark on AWS EMR to process terabytes of transactional and operational data for enterprise reporting and machine learning workflows.
- Designed and implemented ETL workflows using Spark (Java and PySpark), integrating data from Amazon S3, relational databases (PostgreSQL, MySQL), and NoSQL sources (MongoDB) into AWS-based data lakes.
- Built and optimized data warehousing solutions on Amazon Redshift, implementing dimensional modeling (star/snowflake schemas), sort/dist keys, and spectrum queries to support real-time BI reporting.
- Developed AI-driven anomaly detection pipelines using PySpark and SageMaker, identifying abnormal patterns in system logs, transactional anomalies, and suspicious login behavior across distributed applications.
- Designed and orchestrated real-time streaming data pipelines with Apache Kafka and Amazon Kinesis, enabling ingestion of live data into EMR and Redshift for downstream processing and analysis.
- Integrated machine learning models developed in SageMaker into batch and real-time Spark workflows on EMR, facilitating use cases such as fraud scoring, customer churn prediction, and inventory optimization.
- Containerized Spark applications and SageMaker inference scripts using Docker and Kubernetes (EKS), enabling scalable, portable deployment across staging and production environments.
- Implemented data quality frameworks using Great Expectations, validating data schema, null thresholds, and range constraints before loading datasets into Redshift.
- Managed infrastructure as code using Terraform and AWS CloudFormation, provisioning EMR clusters, Redshift instances, IAM roles, and networking components in a secure, repeatable fashion.
- Scheduled and monitored DAG-based workflows using Apache Airflow, automating complex dependencies across AWS Glue, EMR, and Redshift to reduce latency and manual intervention.
- Built custom Python-based monitoring tools for ingestion latency, job failures, and model drift, integrated with AWS CloudWatch and Prometheus for real-time observability and alerting.
- Developed Natural Language Processing (NLP) solutions for log analysis and user feedback categorization, using pre-trained transformer models and Spark NLP on EMR.
- Enforced security best practices and compliance via AWS IAM, Lake Formation, and encryption at rest/in transit across all data layers, supporting regulatory frameworks like GDPR and HIPAA.
- Mentored a cross-functional team of junior data engineers on Spark performance tuning, ML model deployment, and cloud-native data engineering practices using AWS ecosystem tools.
- Collaborated with product managers, architects, and data scientists in Agile teams to design scalable data platforms that power business-

critical dashboards and predictive decision engines.

- Developed CI/CD pipelines for data workflows using AWS Code Pipeline, Terraform, and CloudFormation, automating deployment and infrastructure provisioning.
- Implemented Airflow DAGs to orchestrate data pipelines, scheduling complex dependencies across AWS Glue, Redshift, and EMR for batch and streaming workloads.
- Designed fault-tolerant data ingestion pipelines integrating APIs, PostgreSQL, MySQL, and MongoDB, ensuring high availability and consistency of ingested data.
- Mentored junior engineers on Big Data frameworks, Java-based Spark development, and best practices in data engineering, fostering a culture of continuous learning.

Environment: AWS (S3, Glue, EMR, Redshift, Kinesis, SageMaker, IAM, CloudFormation, EKS, CloudWatch), Apache Spark (Java, PySpark), Kafka, Airflow, Terraform, Docker, Kubernetes, Python, Great Expectations, Prometheus, PostgreSQL, MongoDB, NLP.

EDUCATION:

Trine University | Master of Science (MS) | Information Systems.
JNTU-H University | Bachelor of Technology (B.Tech) | Computer Science Engineering.

TECHNICAL SKILLS:

Programming & Scripting	Python, SQL, Java, Scala, Shell Scripting
Big Data & Analytics	Apache Spark, Hadoop, Kafka, Databricks, Flink, Presto, Snowflake
Cloud Platforms	AWS (Redshift, Glue, S3, Lambda, EMR, Athena), Azure (Data Factory, Synapse, Databricks, Data Lake), GCP (Big Query, Dataflow, Cloud Storage)
Data Engineering Tools	Apache Airflow, dbt, Apache Beam, Talend, Informatica.
Databases	PostgreSQL, MySQL, MongoDB, DynamoDB, Cassandra, Oracle
Streaming & Messaging	Kafka, Kinesis, Pub/Sub, RabbitMQ
Data Warehousing	Snowflake, Redshift, Big Query, Azure Synapse
Visualization Tools	Tableau, Power BI, Looker, Quick Sight
DevOps & Infrastructure	Docker, Kubernetes, Terraform, Ansible

REFERENCE:

Available Upon Request